

MYSQL PROJECT

TOPIC: STUDENT ACADEMIC PERFORMANCE ANALYSIS USING MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of a student academic performance dataset using MySQL. The dataset contains information about students, including gender, race or ethnicity, parental level of education, lunch type, test preparation course, and scores in mathematics, reading, and writing.

The objective of this analysis is to examine student academic performance and identify meaningful patterns and relationships among different student characteristics and their examination scores.

The project uses SQL techniques such as filtering, sorting, aggregation, grouping, conditional analysis, and subqueries to analyze the dataset and generate useful insights from the student performance data.

TOOL: MySQL Workbench

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Queries

1. Display the first 10 student records showing all columns: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score, and writing score.

```
select gender, `race/ethnicity`, `parental level of education`,  
lunch, `test preparation course`, `math score`, `reading score`,  
`writing score` from performance limit 10;
```

	gender	race/ethnicity	math score	reading score	writing score
▶	female	group B	90	95	93
	male	group C	76	78	75
	female	group B	88	95	92
	male	group C	88	89	86
	male	group E	97	87	82
	male	group E	81	81	79
	female	group D	75	90	88
	male	group C	82	84	82
	male	group D	88	78	75
	female	group D	85	91	89

Insight: The first 10 records provide an overview of the student performance dataset. The data contains information about students' gender, race/ethnicity, parental level of education, lunch type, test preparation status, and their scores in mathematics, reading, and writing. The records also show that student performance varies across the three subjects.

2. Calculate the average math score, reading score, and writing score for each gender.

```
select gender, avg(`math score`) as average_math_score,  
avg(`reading score`) as average_reading_score, avg(`writing  
score`) as average_writing_score from performance group by  
gender;
```

	gender	average_math_score	average_reading_score	average_writing_score
▶	female	63.6332	72.6081	72.4672
	male	68.7282	65.4730	63.3112

Insight: The query compares the average performance of male and female students across mathematics, reading, and writing. It helps identify differences in average subject performance between the two gender groups.

- Find the number of students in each test preparation course category and arrange the results from the highest number of students to the lowest.

```
select count(*) as no_of_students, `test preparation course` from performance group by `test preparation course` order by count(*) desc;
```

	no_of_students	test preparation course
▶	642	none
	358	completed

Insight: The result shows the distribution of students based on their test preparation status. The none category contains more students than the completed category, indicating that a larger portion of the dataset did not complete the test preparation course.

- Display 10 students whose math score is 80 or above, showing their gender, race/ethnicity, math score, reading score, and writing score.

```
select gender, `race/ethnicity`, `math score`, `reading score`, `writing score` from performance where `math score` >= 80 limit 10;
```

	gender	race/ethnicity	math score	reading score	writing score
▶	female	group B	90	95	93
	female	group B	88	95	92
	male	group C	88	89	86
	male	group E	97	87	82
	male	group E	81	81	79
	male	group C	82	84	82
	male	group D	88	78	75
	female	group E	82	85	86
	male	group A	80	78	81
	female	group D	85	91	89

Insight: The query identifies students with high mathematics scores and displays their performance in all three subjects. This allows high-performing mathematics students to be compared with their reading and writing performance.

5. Find the highest math score, highest reading score, and highest writing score in the performance table.

```
select max(`math score`) as highest_math_score, max(`reading score`) as highest_reading_score, max(`writing score`) as highest_writing_score from performance;
```

	highest_math_score	highest_reading_score	highest_writing_score
▶	100	100	100

Insight: The query identifies the maximum score achieved in each subject. The result shows that the highest score in mathematics, reading, and writing is 100, indicating that students achieved the maximum possible score in all three subjects.

6. Find the number of students who scored 70 or above in all three subjects: math score, reading score, and writing score.

```
select count(*) as number_of_students from performance where `math score` >= 70 and `reading score` >= 70 and `writing score` >= 70;
```

	number_of_students
▶	324

Insight: The query identifies students who maintained a score of at least 70 across all three subjects. This helps measure how many students demonstrated consistently good academic performance rather than performing well in only one subject.

7. Display 10 students whose reading score is higher than their math score.

```
select gender, `race/ethnicity`, `math score`, `reading score`  
from performance where `reading score` > `math score` limit  
10;
```

	gender	race/ethnicity	math score	reading score
▶	female	group C	69	90
	female	group B	90	95
	male	group A	47	57
	male	group C	76	78
	female	group B	71	83
	female	group B	88	95
	male	group B	40	43
	female	group B	38	60
	male	group D	40	52
	female	group B	65	81

Insight: The query identifies students whose reading performance was better than their mathematics performance. This provides a way to compare subject-specific strengths within individual student records.

8. Find the average math score for each race/ethnicity group and display only groups where the average is greater than 65

```
select avg(`math score`) as average_math_score,  
`race/ethnicity` from performance group by `race/ethnicity`  
having avg(`math score`) > 65;
```

	average_math_score	race/ethnicity
▶	67.3626	group D
	73.8214	group E

Insight: The query compares average mathematics performance across different race/ethnicity groups and filters out groups with an average score of 65 or below. This helps identify groups with relatively higher average mathematics performance.

9. Display 10 students whose parental level of education is either bachelor's degree or master's degree.

select gender, `race/ethnicity`, `parental level of education`, `math score` from performance where `parental level of education` in ("bachelor's degree", "master's degree") limit 10;

	gender	race/ethnicity	parental level of education	math score
▶	female	group B	bachelor's degree	72
	female	group B	master's degree	90
	female	group A	master's degree	50
	male	group C	master's degree	46
	male	group D	bachelor's degree	74
	male	group A	master's degree	73
	female	group C	bachelor's degree	67
	female	group D	master's degree	62
	female	group E	master's degree	56
	male	group E	bachelor's degree	79

Insight: The query focuses on students whose parents have higher levels of formal education, specifically a bachelor's or master's degree. It allows their demographic information and mathematics performance to be examined.

10. Find 10 students whose math score is higher than the overall average math score.

select gender, `race/ethnicity`, `math score`, `reading score`, `writing score` from performance where `math score` > (select avg(`math score`) from performance) limit 10;

	gender	race/ethnicity	math score	reading score	writing score
▶	female	group B	72	72	74
	female	group C	69	90	88
	female	group B	90	95	93
	male	group C	76	78	75
	female	group B	71	83	78
	female	group B	88	95	92
	male	group A	78	72	70
	female	group C	69	75	78
	male	group C	88	89	86
	female	group C	69	73	73

Insight: The subquery first calculates the overall average math score. The main query then identifies students whose mathematics score is above that average. This helps distinguish students performing above the overall mathematics performance level in the dataset.

11. Find the lowest math score, lowest reading score, and lowest writing score in the performance table.

```
select min(`math score`) as lowest_math_score, min(`reading score`) as lowest_reading_score, min(`writing score`) as lowest_writing_score from performance;
```

	lowest_math_score	lowest_reading_score	lowest_writing_score
▶	0	17	10

Insight: The query identifies the minimum score achieved in each subject. It provides a quick view of the lowest recorded performance in mathematics, reading, and writing.

12. Display 10 students whose math score is between 60 and 70, inclusive.

```
select gender, `race/ethnicity`, `math score`, `reading score` from performance where `math score` between 60 and 70 limit 10;
```

	gender	race/ethnicity	math score	reading score
▶	female	group C	69	90
	male	group D	64	64
	female	group B	65	81
	female	group C	69	75
	male	group D	66	69
	female	group B	65	75
	female	group C	69	73
	male	group B	69	54
	female	group C	67	69
	male	group C	70	70

Insight: The query identifies students whose mathematics scores fall within the range of 60 to 70. This helps examine students performing within a moderate mathematics score range.

13. Display 10 students whose parental level of education is not high school.

select gender, `race/ethnicity`, `parental level of education`, `math score` from performance where `parental level of education` <> 'high school' limit 10;

	gender	race/ethnicity	parental level of education	math score
▶	female	group B	bachelor's degree	72
	female	group C	some college	69
	female	group B	master's degree	90
	male	group A	associate's degree	47
	male	group C	some college	76
	female	group B	associate's degree	71
	female	group B	some college	88
	male	group B	some college	40
	male	group C	associate's degree	58
	male	group D	associate's degree	40

Insight: The query filters out students whose parental education level is high school and displays students with other educational backgrounds. This allows their mathematics performance to be examined across different parental education levels.

14. Find 10 students whose math score is greater than their reading score, and whose reading score is greater than their writing score.

select gender, `race/ethnicity`, `math score`, `reading score`, `writing score` from performance where `math score` > `reading score` and `reading score` > `writing score` limit 10;

	gender	race/ethnicity	math score	reading score	writing score
▶	male	group C	58	54	52
	male	group A	78	72	70
	male	group E	97	87	82
	male	group E	77	69	68
	male	group C	53	44	42
	male	group D	88	78	75
	male	group E	79	74	72
	male	group A	62	61	55
	male	group B	67	64	61
	male	group B	61	58	56

Insight: The query identifies students whose scores follow a specific descending pattern from mathematics to reading to writing. This helps identify students with a consistent difference in performance across the three subjects.

15. Find 10 students who scored at least 75 in both math score and reading score, but whose writing score is lower than their reading score.

```
select gender, `race/ethnicity`, `math score`, `reading score`,  
`writing score` from performance where `math score` >= 75  
and `reading score` >= 75 and `writing score` < `reading  
score` limit 10;
```

	gender	race/ethnicity	math score	reading score	writing score
▶	female	group B	90	95	93
	male	group C	76	78	75
	female	group B	88	95	92
	male	group C	88	89	86
	male	group E	97	87	82
	male	group E	81	81	79
	female	group D	75	90	88
	male	group C	82	84	82
	male	group D	88	78	75
	female	group D	85	91	89

Insight: The query identifies students who performed well in mathematics and reading but had a comparatively lower writing score. This helps highlight differences in subject-wise performance among students with generally good scores.

CONCLUSION

The **Student Performance Analysis** project successfully demonstrated the use of **MySQL** to analyse student academic data. Using the performance table, various SQL techniques such as filtering, aggregation, GROUP BY, HAVING, IN, BETWEEN, logical conditions, aggregate functions, and subqueries were applied to extract meaningful information from the dataset.

The analysis provided insights into students' performance in **math score**, **reading score**, and **writing score**, along with factors such as gender, race/ethnicity, parental level of education, and test preparation course. The queries helped identify average and highest scores, students with consistently good performance, score comparisons between subjects, and students performing above the overall average.

Overall, this project strengthened practical knowledge of **SQL** and **MySQL Workbench** and demonstrated how raw student data can be transformed into useful information through structured SQL queries. The analysis also showed how SQL can be used to identify patterns and compare academic performance across different groups.